

Ashin Murugan

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SUMMARY

M.Tech Cyber Security student with a strong foundation in cloud platforms (AWS, GCP) and hands-on experience securing government networks. Practical background in conducting security assessments, identifying vulnerabilities, and translating complex technical problems into clear solutions for both engineering teams and business leaders. Passionate about applying analytical skills to cloud security, cloud observability, competitive research, system automation and infrastructure defense.

EDUCATION

Vellore Institute of Technology

Integrated M.Tech – Computer Science and Engineering (Cyber Security)

CGPA: 8.74 / 10

Oct 2022 – Present

TECHNICAL SKILLS

- **Application Security:** Web Security, OWASP Top 10, Secure Code Review, VAPT, Threat Modeling, Security Controls Analysis
- **Security Tools:** Burp Suite, Nmap, Wireshark, Metasploit, OWASP ZAP, Wazuh SIEM, Zabbix, MS Sysinternals, Linux
- **Cloud & Platforms:** AWS (IAM, EC2, S3, VPC, Lambda), GCP, Cloud Security Architecture, SIEM Integration, Zero Trust
- **Scripting & Development:** Python, FastAPI, React, MySQL, Git
- **Networking & Forensics:** TCP/IP, Packet Analysis, Network Defense, SOC Operations, Incident Response, MITRE ATT&CK, Cryptographic Protocols

EXPERIENCE

High Court of Kerala

Ernakulam, Kerala, IN

Cyber Security Intern

Oct 2025 – Dec 2025

- Conducted security assessments across 47+ endpoints in a regulated government environment; identified and remediated 18+ critical vulnerabilities aligning with strict compliance mandates.
- Synthesized technical findings into actionable reports, bridging raw vulnerability data and IT management strategy to improve architectural defense.
- Wrote Python scripts to automate log collection and analysis, cutting manual review time by 40% and improving consistency across compliance checks.
- Partnered with the engineering team to architect and deploy a Unified Ops Dashboard (Wazuh and Zabbix), providing centralized visibility into distributed system events.

PROJECTS

Modified Dynamic Block Cipher

[GitHub](#)

Cryptography, Feistel Network, Python

Oct 2023 – Nov 2023

- Designed a 32-round Generalized Feistel Network (GFN) cipher with a 4-branch architecture, enhancing data diffusion and entropy beyond traditional 2-branch designs.
- Implemented a dynamic key schedule from a 1024-character space with an 8-bit to 64-bit binary expansion mapping to ensure a high avalanche effect and prevent frequency analysis.

Unified Ops Security Monitoring Dashboard

[GitHub](#)

Python, FastAPI, Wazuh SIEM, Zabbix

Oct 2025 – Dec 2025

- Architected a centralized SOC dashboard integrating Wazuh SIEM and Zabbix across 13+ distributed servers; reduced alert latency to under 15 seconds for suspicious activity.
- Configured Prometheus to scrape performance and security metrics from cloud workloads, developing custom Grafana dashboards for proactive incident detection.
- Secured the backend with end-to-end encrypted communication; system handles 1,000+ metrics per minute reliably.

AgentWall – Intelligent AI Security Layer

[GitHub](#)

Python, FastAPI, React, DuckDB, psutil

Apr 2026 – May 2026

- Engineered a security layer using heuristic process monitoring to correlate OS-level file modifications with autonomous AI agent processes (Cursor, Aider, VS Code) in real-time.
- Implemented an asynchronous telemetry pipeline with DuckDB and HMAC-chained logging, handling 100+ security events per second with cryptographic log integrity and zero latency impact.
- Developed a forensic SOC dashboard visualizing MITRE ATT&CK technique mapping with step-by-step session replay for LLM-based tool interactions and system-level modifications.

ACHIEVEMENTS

- **Research Publication (ICSTSDG 2024):** Co-authored a paper on automated security surveillance using computer vision in public spaces.
- **Amazon ML Challenge 2025:** Ranked 192nd globally out of 8,000+ participants.
- **Conference Paper (IC-ICNS 2026):** Presented CT-GAN, a Conditional Transformer GAN for generating realistic phishing emails to support adversarial testing and threat intelligence.

CERTIFICATIONS

- **Amazon Web Services:** AWS Certified Cloud Practitioner
- **Google Cloud:** Google Cloud Cybersecurity Certificate (*Beginner*)
- **Cisco Certified:** Ethical Hacker, Network Defense **Google:** The Bits and Bytes of Computer Networking